No Feature Selection Algorithm Rules Supreme, but Some Yield Better Classifiers

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Abstract—Feature selection is the process of selecting a subset of features based on data. There are many feature selection algorithms available and the appropriate choice can be difficult. The aim of this paper was to benchmark algorithms in order to provide guidance for this choice. First, we assessed which algorithms are most commonly used in the scientific community by performing a literature study in two large reference databases: Scopus and Web of Science. Second, we constructed a benchmark pipeline to compare the performance of 31 algorithms on 50 datasets. The feature subset was reduced to 50 % of the original size in all cases. Exploration of dataset characteristics and their effects on the experiment was beyond the scope of this study. The algorithm performance was measured based on classification and computational time. No algorithm was superior in every case. The overall best-performing algorithm was Decision Tree as an embedded feature selection method. Lasso also showed good predictive performance and speed. Embedded versions of Random Forest and AdaBoost yielded the best average predictive performance, but were significantly slower.

Index Terms—Machine learning, Data mining, Data processing, Classification, Classifier evaluation, Benchmark, Comparison, Performance evaluation of algorithms, Feature evaluation and selection, Attribute selection, Variable selection, Dimensionality reduction, Model selection, Predictive performance, Recursive feature elimination, Sequential forward selection, Sequential backward selection, Decision Tree, Random Forest, AdaBoost, ANOVA, CFS, Chi Squared, FCBF, LARS, LASSO, Linear Regression, Low Variance Filter, MRMR, Perceptron, ReliefF, SVM

1 Introduction

Datasets are growing, both in the number of observations and dimensions (Fig 1). More data means greater potential for discoveries, predictions and understanding, but must be processed in order to be useful [1]. One preprocessing approach is determining what information best displays the underlying mechanisms of the observed phenomenon.

Feature selection is a process that automatically selects useful subsets of features from datasets. This removes noise which reduces computation costs and improves both predictive performance and model interpretability [3]. The importance of feature selection in statistics, data mining, model building and machine learning is undisputed, and the importance will increase as datasets grow [4], [5], [6].

There are many *feature selection algorithms* (FSAs) available, but knowing which to use is not easily discerned [6]. Several studies show that the performance of different FSAs varies significantly when applied on the same data [1], [7], [8]. Some have also showed that incorrectly applied feature selection can reduce predictive performance [1], [9].

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Many benchmarks have been done, yet how to choose the best FSA remains an open research question. The problems with many existing benchmarks are:

- Poor or no motivation for the choice of algorithms, and
- too few algorithms and datasets.

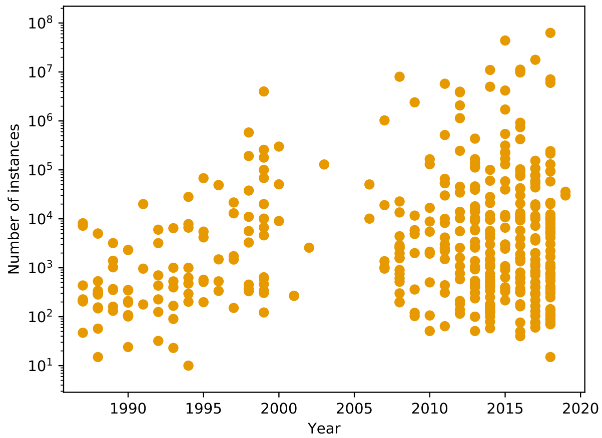
The aim of this paper is therefore to answer: Which feature selection algorithm(s) among those that are prevalent in the scientific literature has the overall best performance?

2 Theory

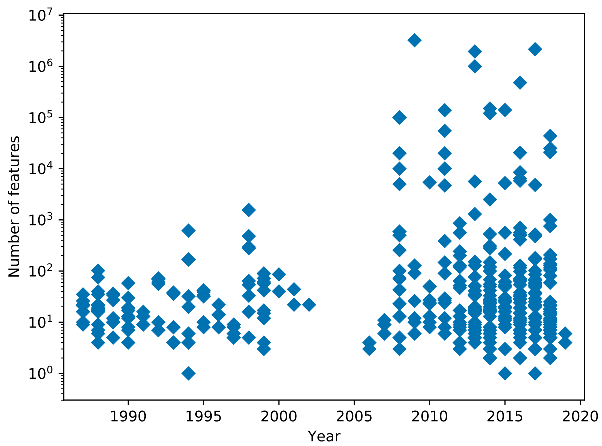
Feature selection is a data preprocessing method usually done before training a learner. The performance of FSAs are typically measured implicitly by evaluating the predictive performance of the learner. FSAs can also have internal learners, and some learners carry out internal feature selection as part of the training process. Definitions of FS vary across studies, depending on which aspect of the process the researcher emphasizes. We therefore gathered 28 definitions from 24 different papers, that were then rewritten as mathematical optimization problems (Appendix A). The identified overlap resulted in a proposed general definition which is used for this paper:

Feature selection is the process of including or/and excluding some features or data in modelling in order to achieve some goal, typically a trade-off between minimising the cost of collecting the data and fulfilling a performance objective on the model.

There are several reasons for omitting features. If they have low correlation to the class label the model can become too complex and represent noise and random error rather than underlying causality. This can lead to bad predictive



(a)



(b)

Fig. 1. The increase of size (a) and dimensionality (b) of datasets. Data from UCI Data Set Repository [2]. Note that the number of datasets in this repository is increasing; 8 datasets were uploaded 1987 compared to 51 in 2018.

performance since the model is overfitted and therefore generalises poorly on unseen data [3]. Domingos [10] has called overfitting the most important problem in machine learning. Another reason for feature selection is better learning efficiency; simpler datasets means faster learning. Additionally, more features requires many more instances for the learner to find the best solution, since the density of the instances decreases exponentially as the dimensionality increases. Reducing the number of features is thus tantamount to increasing the number of instances [3].

The categories commonly used to distinguish FSAs are *wrapper*, *embedded* and *filter*, however there are discrepancies between the definitions. The descriptions below were formulated after analysis of eight papers that presented definitions [1], [3], [5], [6], [11], [12], [13], [14].

2.1 Wrapper

The features are selected by a combination of search strategy and learner. The learner is trained with various feature subsets and the one that yields the best predictive performance is kept. Consider a dataset with three features $D = \{x_1, x_2, x_3\}$. Three subsets, $S_1 = \{x_1, x_2\}, S_2 = \{x_1, x_3\}, S_3 = \{x_2, x_3\}$ are separately used to train a learner and test its performance with some objective function $J(\cdot)$ which is to be optimised. If $J(S_3) > J(S_1) > J(S_2)$, the selected features are x_2 and x_3. However the only way to guarantee optimal performance requires testing all seven feature combinations with an exhaustive search [1], which is only feasible if the number of features is small [1], [15]. Heuristic search strategies enables finding adequate solutions in a feasible time frame by only testing some of all possible combinations.

The search strategies we used were *Sequential Forward Selection* (SFS) and *Sequential Backward Selection* (SBS). The former starts with an empty feature set and sequentially adds the features that yield the best performance, until a desired subset size is reached. The latter starts with a full feature set and sequentially removes the features that yield the worst performance. Both strategies have the same complexity but generate different candidate subsets, which can lead to different solutions. The names of the wrappers in this paper is a combination of the name of learner and search strategy, for example Decision Tree SFS, Random Forest SFS, and Support Vector Machine (SVM) SBS.

2.2 Embedded

Learners can also be used for feature selection by examining model structure instead of performance. Consider a case where the trained learner is a linear regression model. The model is represented by the function y, $\bar{x}$ is the feature vector and $\bar{C}$ is a vector of coefficients:

$$y(\bar{x}) = \bar{C}\bar{x} = C_1x_1 + C_2x_2 + C_3x_3$$

Assuming that all features have been normalised to same norm, the importance of a feature is indicated by its corresponding coefficient. Unimportant features are omitted by choosing some length of $\bar{x}$ or threshold value of $\bar{C}$. If for example we want to omit one feature and $C_2 > C_3 > C_1$, the selected subset will be $S = \{x_2, x_3\}$.

We included two different types of embedded FSAs in the experiment. The type where the learner is trained once, as in the example above, is named *standard embedded* in this paper. The other type is *Recursive Feature Elimination (RFE)* [16], which iteratively re-trains the learner after removing the least important feature. Since both types can involve the same learner, the embedded FSAs in this paper are titled by the learner used, with the suffix 'RFE' added to those that use recursive feature elimination, for example Lasso, AdaBoost, AdaBoost RFE.

2.3 Filter

Feature selection without learners can be done by measuring intrinsic data properties. If for example a feature has the same value over all instances, it is useless for discerning between class labels. The relationship to the class label can be estimated by calculating test statistics, such as F-test or χ^2. Some filter methods, such as *minimum Redundancy Maximum Relevance (mRMR)* and *Correlation-based Feature Selection (CFS)* also measure feature-to-feature correlation in order to remove redundant features.

TABLE 1
Summary of FSA categories.

	Involves learners	Basis of selection	Ranking or subset selection
Wrapper	Yes	Model performance	Subset
Embedded	Yes	Model structure	Ranking
Filter	No	Statistical measures	Ranking

2.4 Feature ranking or subset selection

Another way to categorise FSAs is if they select features individually or in groups (Table 1). Embedded and filter methods rank features according to their individual importance. Wrappers select entire subsets without discerning which features in the subset are most relevant. Embedded and filter methods should therefore be faster, especially when dealing with high-dimensional data [17]. Some have even claimed their speed scales linearly with the number of features [18]. Wrappers on the other hand, excel in situations where a combined set of features is optimal regardless of their individual importance.

3 Method

3.1 Inclusion criteria

The biggest criterion for inclusion of a FSA was prevalence in published papers. To uncover as many publications as possible it was necessary to use a comprehensive search string. For this purpose, the list compiled by Spolaôr et al. [19] was modified and extended to a full set of terms (Appendix B). This search string was used to collect review articles from Scopus and Web of Science, from which a total of 102 FSA names were gathered. Every abbreviation and synonym for these FSAs were then used in combination with the search string in Scopus and Web of Science. For example, the input for mRMR was *("mrmr" OR "minimum redundancy maximum relevance" OR "maximum relevance minimum redundancy") AND ("feature selection" OR "variable selection" OR "attribute selection" OR ...".* The number of hits acted as an approximation of a FSA's prevalence. The top results of these searches are made available in Appendix C.

The remaining criteria was availability and variety. We relied on freely available software packages (Appendix D) and included different types of FSAs. Ultimately, a set of 31 FSAs were included in the benchmark (Table 2). Analogously, the external learners were also chosen to have different underlying mathematics. Inspiration was taken from the work of Wu et al. [20], which attempted to identify the top ten learners used in data mining. Gaussian Naive Bayes, K-Nearest Neighbour and Decision Trees, were finally chosen for evaluation of the FSAs. To avoid selection bias due to a particular choice of datasets, the comparison was done using 50 well-known publicly available datasets from different scientific fields. Most datasets were derived from real world measurements with varied number of instances, features and classes, but upper limitations were imposed due to computational costs (Appendix E).

TABLE 2
The 31 FSAs included in the experiment

Name	Category	Source
AdaBoost AdaBoost RFE	Embedded	[21]
AdaBoost SBS AdaBoost SFS	Wrapper	
Anova	Filter	[22]
Correlation Based FS (CFS)	Filter	[11]
Chi Squared	Filter	[23]
Decision Tree Decision Tree RFE	Embedded	[24]
Decision Tree SBS Decision Tree SFS	Wrapper	
Fast Correlation Based FS (Fcbf)	Filter	[25]
Least Angle Regression (Lars)	Embedded	[26]
Least Absolute Shrinkage and Selection Operator (Lars)		[27]
LassoLars		[26]
Linear Regression		-
Low Variance	Filter	[22]
Minimum Redundancy Maximum Relevance (mRMR)	Filter	[28]
Perceptron Perceptron RFE	Embedded	[29]
Perceptron SBS Perceptron SFS	Wrapper	
Random Forest Random Forest RFE	Embedded	[30]
Random Forest SBS Random Forest SFS	Wrapper	
ReliefF	Filter	[31]
Support Vector Machines (SVM)* Support Vector Machines (SVM)* RFE	Embedded	[32]
Support Vector Machines (SVM)** SBS Support Vector Machines (SVM)** SFS	Wrapper	

*Linear kernels, **Nonlinear kernels
RFE– Recursive Feature Elimination
SBS– Sequential Backward Selection
SFS– Sequential Forward Selection

3.2 Experimental Setup

The predictive performance of the FSAs were measured using the three aforementioned external learners, with accuracy and F-measure as scoring metrics. Feature selection, classification and hyperparameter optimisation (HPO) for both were performed within each fold of a stratified 5-fold cross validation sampling scheme on each of the 50 datasets. The HPO consisted of an exhaustive grid search of the parameters listed in Appendix F, which also includes the tuning options for the external learners. Most FSAs also required a choice of subset size. To ensure a fair comparison, subsets from all FSAs were restricted to the same size, which was 50% of the original feature vector. After averaging over the external learners, the result for each

TABLE 3
The best performing FSAs. The types of FSA are specified: Embedded (E) and Filter (F). Bold FSA are present in both groups. Italicised FSAs are present in only one of the groups, but have noteworthy high comparative performance in the other

Predictive performance		**Runtime**	
FSA	**Type**	**FSA**	**Type**
Random Forest RFE	E	Low Variance Filter	F
Random Forest	E	**Decision Tree**	E
Decision Tree RFE	E	Anova	F
Decision Tree	E	Chi Squared	F
AdaBoost RFE	E	*Lasso*	E
AdaBoost	E	Perceptron	E
		LassoLars	E

FSA and performance metric consisted of one crossvalidated score for further analysis. For reference, performance scores for the external learners without FS was included in the comparison, denoted as *NoFS*.

Speed was also included as an additional performance metric in the experiments. If an algorithm is fast, it allows for more scrupulous parameter tuning which can enhance predictive performance. We therefore chose to record the average time for each FSA to produce a subset. Since we were going to compare FSAs across multiple datasets of various sizes, the runtimes were rescaled. The rescaled value x' ranges between 0 (slowest) to 1 (fastest) using x as the initial runtime and x_{min} and x_{max} as the largest and smallest recorded runtimes respectively:

$$x' = 1 - \frac{x - x_{min}}{x_{max} - x_{min}}$$

4 Results

The best performing group of FSAs is presented in Table 3. Both accuracy and F-measure shared the same best performing group, all of which were of embedded types. With addition to SVM RFE, this group consisted of the only FSAs that performed better than omitting feature selection entirely (named ”noFS” in the resulting graphs); the difference in performance of remaining 24 FSAs were not statistically significant at α 0.05 to that of the external learners alone. Runtimes revealed a different group, which included both filter and embedded methods. For overall performance, one particular FSA appeared in both of the best performing groups: Decision Tree. Two FSAs were present among the best performing within one measure while also having noteworthy high comparative performance in the other: Decision Tree RFE and Lasso.

4.1 Statistical analysis

The analysis consisted of statistical hypothesis testing with a method accounting for the distribution of the results and the experimental setting, which involved multiple datasets. The distribution for all three metrics was assessed using Tukey boxplots (Appendix G). All showed skewness and modality, which implied a non-normal distribution and warranted a non-parametric statistical method for significant comparison. We therefore chose to use an extension to the Friedman test derived by Iman and Davenport (1980) [33] with the Nemenyi post-hoc test as implemented by Demšar (2006) [34].

Both accuracy and F-measure displayed similar ordering of FSAs regarding predictive performance and ultimately shared the best performing group. The calculated F_F statistic was $F_F \approx 27$ and depended on $k = 32$ number of FSAs and $N = 50$ number of datasets, including the result for just the external learner, depicted as No FS. The tabular critical F-value for $F(31, 1519) = 1.46$, and since $F_F > F(31, 1519)$, non-equivalence was shown in predictive performance between FSAs at significance level $\alpha = 0.05$. For finding which FSA or groups of FSAs that differ, the Nemenyi critical distance (CD) was calculated using the same k and N and the tabular Nemenyi critical value $q_{0.05} = 3.78$, resulting in $CD_{0.05} \approx 7.09$. The performance of two FSAs are considered different if their average ranks R_j differ by at least $CD_{0.05}$, shown as FSAs with non-overlapping bars in Figure 2. The figure shows the top-performing FSA regarding predictions was Random Forest RFE. This FSA was grouped together with the subsequent five algorithms since it was not separable ($\alpha = 0.05$) from these and are presented together as the best performing group in Table 3 in the summary. The figure also shows that apart from the aforementioned group of six FSAs and SVM RFE, the performance of the remaining 24 FSAs were not significantly different from NoFS at significance level $\alpha = 0.05$.

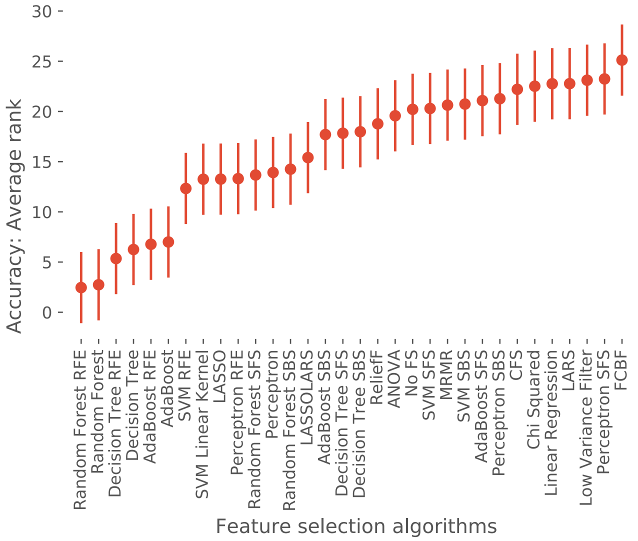
Analogous to the analysis of predictive performance, the distributions of rescaled runtimes from slowest to fastest (0-1) for each FSA on each dataset were first inspected with raw data overlaying Tukey boxplots (Appendix G). The distributions strongly implied nonnormailty, so we again used the Iman Davenport Friedman test and subsequently the Nemenyi post-hoc test for elucidating the differences.

With $k = 31$ number of FSAs and $N = 50$ number of datasets, the calculated F_F statistic was $F_F \approx 89.62$. The tabular critical F-value for $F(30, 1470) = 1.47$, and since $F_F > F(30, 1470)$, the runtimes can be stated as nonequivalent ($\alpha = 0.05$). The following Nemenyi post-hoc test resulted in $CD_{0.05} \approx 6.85$ using previously defined k and N and the tabular critical value $q_{0.05} = 3.76$. The speed of two FSAs can be considered significantly different if their corresponding average ranks differ by at least the calculated $CD_{0.05}$, which translates to FSAs with non-overlapping bars in Fig. 3. The fastest FSA was shown to be Low Variance Filter. Since the following six algorithms could not be distinguished as significantly ($\alpha = 0.05$) different , they together formed the group of the fastest FSAs presented in the summary (Table 3)

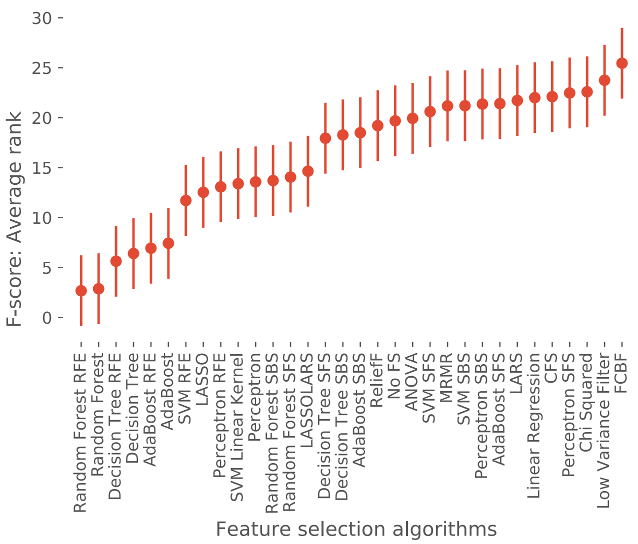
4.2 Discerning best overall performance

The overall best performing FSAs were those that achieved the best average ranks with respect to both predictive performance and speed. Consequently, plots presenting CD for accuracy, F-measure (Fig. 2) and runtime (Fig.3) from the previous section were compared, considering only the best performing group and the consecutive statistically indistinguishable group.

As shown in Fig. 4, Decision Tree was present in the best performing group with respect to predictive performance



(a)



(b)

Fig. 2. Comparison of estimated accuracies (a) and F-measures (b) between FSAs across all datasets. The dots represent the average ranking of each FSA, with upper and lower boundaries representing the Nemenyi critical distance. The FSAs are ordered from lowest to highest average rank $(+/-0.5CD_{0.05})$. Groups with non-overlapping bars are considered significantly different $(\alpha = 0.05)$.

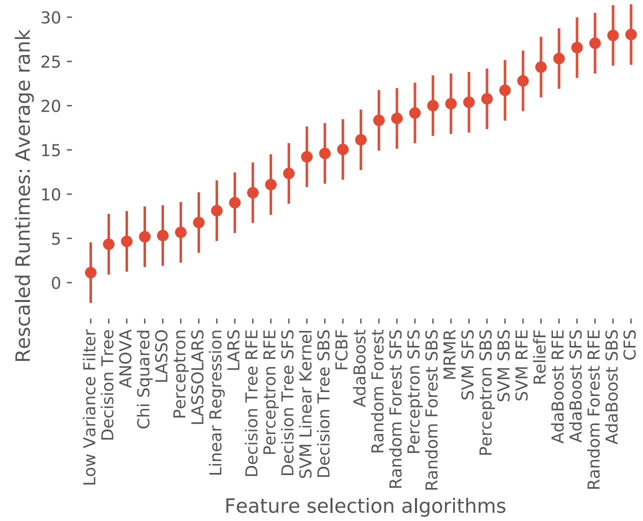


Fig. 3. Comparison of estimated and rescaled runtimes between FSAs across all datasets. Dots represent the average rank (AR) of each FSA across datasets, with upper and lower boundaries together representing the Nemenyi critical distance (CD). The FSAs are ordered from lowest (best performance) to higher (worst performance) AR $(+/-0.5CD_{0.05})$. Groups with non-overlapping bars are considered significantly different $(\alpha = 0.05)$.

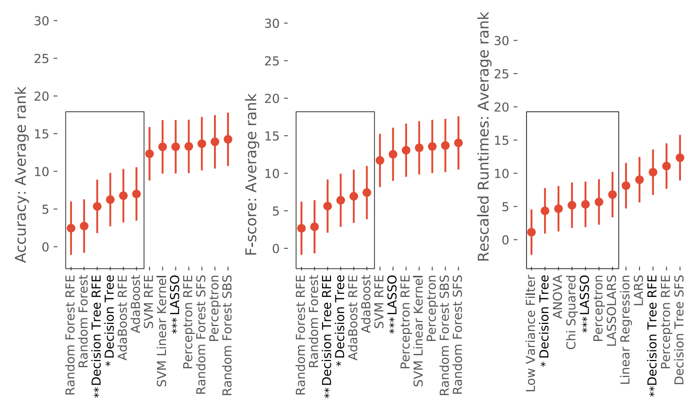


Fig. 4. Discerning FSAs with best overall performance. Plots of average ranks with Nemenyi critical distances are used for accuracy (Fig.2a), F-measure (Fig.2b) and runtime (Fig.3) from earlier sections. Only the best performing group (within rectangle) and the consecutive, statistically indistinguishable group of FSAs are compared. Decision Tree (*) was present in all the best performing groups of FSAs. Decision Tree RFE (**) and Lasso (***) were present in a best performing group with regards to one measure while simultaneously present in the consecutive group of the other.

and runtime, respectively. Decision Tree RFE was in the best performing group regarding prediction, while the difference was not significant ($\alpha = 0.05$) from the majority of best performers regarding runtime. Analogously, Lasso was present in the best performing group regarding runtime, while not significantly different from the majority in the best performing group regarding predictive performance. Decision Tree, Decision Tree RFE and Lasso are therefore emphasized in the summary Table 3 as the FSAs with top overall performance in the experiments.

5 Discussion

5.1 Top performers

The experiment produced two best-performing groups regarding prediction and speed, of which the majority were embedded types. Regarding predictive performance, it was notable that all methods involved learners that use tree structures: Decision Tree, Random Forest and AdaBoost (Table 3). The respective wrapper versions of these were also top-ranked among wrapper types. The predictive performance of these methods was not significantly different but Decision Tree RFE was faster (Fig. 4). Since Decision Tree was the only one present in both best performing groups, it can be considered the best overall performing FSA.

Apart from a few filter types, the majority of the best performing FSAs regarding speed were also embedded types. Lasso in particular was found to be among the fastest FSAs while also having good predictive performance (Fig. 4). However, generalisations based on FSA types should be viewed with caution, since the number of wrapper, embedded and filter methods are unbalanced.

5.2 Runtime

We did not expect any of the slowest FSAs to be filter types. Filter methods that only assess feature-to-label correlation (Low Variance Filter, Anova, Chi Squared) were among the fastest. However, two filter methods that also assess feature-to-feature correlation (ReliefF, CFS) were among the slowest. It seems that calculating redundancy can take a longer time than training learners.

Several learners were used in both wrapper and embedded FSAs and we expected their relative speed to be, from fastest to slowest: embedded, RFE, SFS, SBS. The order is reasonable since it reflects the number of training cycles and the feature vector sizes within each cycle. Standard embedded carries out only one cycle while the number of cycles for RFE depends on the size of the original vector. Both wrapper types, on the other hand, carry out a similar number of cycles but SFS starts with one feature and sequentially increases the number; SBS starts with all features and sequentially decreases the number. If allowed to perform full searches, the two should be equally fast, but in our benchmark we imposed a fixed 50 % restriction on subset size. SBS thus becomes slower than SFS since it trains on larger feature vectors. Both SBS and SFS execute many more training cycles than the embedded types. It was therefore surprising that the FSAs based on Random Forest and SVM had a different ordering, with some wrapper types faster than embedded: embedded, SFS, SBS, RFE for Random Forest, and embedded, SFS, SBS, RFE for SVM (Fig. 3). This indicates that for these algorithms, model construction takes less time than extracting feature importance from model structures.

The results show that filter and embedded FSAs can be slower than wrappers. Most papers consistently claim the opposite [5], [6], [13], [14], [17], [35], [36].

5.3 Predictive performance

We believe the fixed restriction on subset sizes had large implications on the predictive performances. If a dataset could have been sufficiently represented by less than half of the features, noisy features would have been retained and degraded performance. Conversely, the performance would also have been hurt if more than half of the features were optimal. It was therefore remarkable that no FSA performed significantly worse than not performing FS at all (Fig. 4), even though the latter allows for all features to be used for prediction. Considering the crude HPO and the fixed subset size, this demonstrates the utility of FS even under suboptimal conditions. Regardless, the subset size restriction could have especially disadvantaged wrapper types since they examine feature combinations instead of individual importance. Other search strategies than SBS and SFS might have handled this better.

A noteworthy observation is that the predictive performances were so similar – since only six FSAs could be discerned from not performing feature selection at all – despite many different kinds of FSAs and datasets involved. This can also be seen by the similar means and medians in Fig.1 in Appendix G. We believe this is due to a combination of the following reasons:

- There could be larger differences in predictive performance, but since subset sizes were fixed at 50% it lowered the likelihood of FSAs choosing very different subsets. If this is the case, adding more datasets would increase the differences in performance.
- The various strengths and weaknesses of the FSAs are evened out across the datasets. Our starting assumption when comparing FSAs on many datasets was: if algorithm A has slightly better performance than algorithm B, this difference will accentuate in proportion to how many times A and B are compared. Paradoxically, considering the No Free Lunch Theorem [37], this assumption is false and adding more datasets would lower the difference.

We have two suggestions for dealing with the aforementioned points. The first point can be solved by not imposing a fixed limit on subset size. It can instead be treated as a hyperparameter to be optimised and the results from all subsets are saved for later comparison. A way to address the second point could be to assess performance in relation to data characteristics. These could be simple such as the number of instances and features, or feature-to-instance-ratio. They could also be more intricate such as mean skewness or class entropy. Wang et al. [35], suggest a series of dataset characteristics on which to base a FSA recommendation method. The raw data generated from our benchmark can be used to explore such an idea.

For the analysis, we chose a non-parametric omnibus method for statistical hypothesis testing. A more powerful parametric test would have probably revealed greater differences and discriminated more FSAs from each other. However, non-normality such as modality and skewness was apparent in the side-by-side Tukey box plots (Appendix G) of the predictive performance. Likewise, the runtime results displayed very unequal variances between FSAs, which is a requirement for most parametric tests [38].

5.4 Related research: other FSA benchmark studies

Several published papers have carried out FSA benchmarks using a strategy similar to ours (Table 4), but the various problems and perspectives differ between the studies. Many researchers examine FSAs for specific implementations and the choice of datasets reflects this.

An alternative strategy is to use synthetic data where all feature importances are known in advance [14]. Instead of testing predictive performance, these experiments measure how many relevant, irrelevant, and redundant features have been selected. This has the advantage of testing FSAs

TABLE 4
Comparison of FSA benchmarks.

Number of datasets	Number of FSAs	Number of Learners	Performance Measures	Source
50	**31**	**3**	**Accuracy, F-measure, Runtime**	**This study**
30	25	3	Accuracy	[5]
7	8	1	AUC	[9]
14	6	2	Accuracy	[8]
115	22	5	Accuracy	[35]
7	2	2	Accuracy	[39]
1	6	1	ROC, F-measure	[40]
11	13	4	Accuracy	[6]
3	5	4	Accuracy	[41]
1	12	0	MMCE*	[42]
2	5	2	Accuracy	[43]
12	8	2	Accuracy	[44]
1	6	5	Accuracy	[45]
3	36	0	AUC, F-measure, Precision	[46]
13	10	2	Accuracy	[47]
2	5	5	AUC	[48]
2	8	2	Accuracy	[49]
1	5	1	Accuracy	[50]

* Mean MisClassification Error

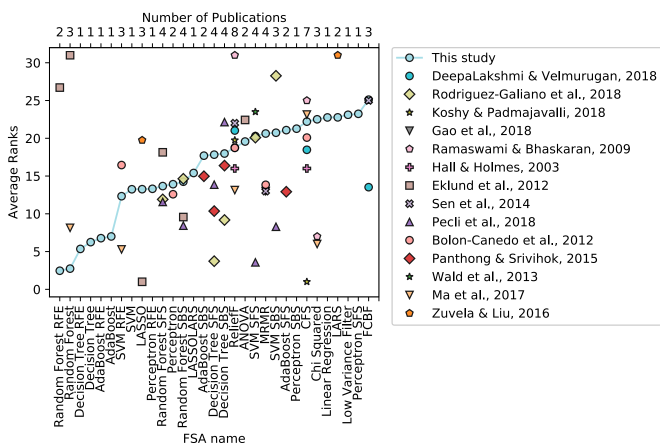


Fig. 5. Average ranks of FSAs according to different comparison studies. The ranks have been rescaled to fit the rank range of our study (1-31). Not many of the studies shared many FSAs with us (≤ 6). Little or no correlation can be observed between rankings of our benchmark and from other studies.

without a layer of obfuscation added by learners. On the other hand, synthetic data does not necessarily reveal performance on real world data, since the structure and noise is difficult to mimic.

Observing various FSA benchmark results (Fig. 5) reveal that comparison of performance between studies is infeasible. There are too many variables affecting the outcome such as the choice of: data, hyperparameter settings, external classifiers, random seeds, performance metrics, sampling scheme and which FSAs to include. Furthermore, concerns have been raised about the lack of standard practices regarding fair comparison of machine learning algorithms (and in extension, feature selection) and that evaluation methods can be biased, misleading and invalid [38], [51], [52], [53]. This necessitates the availability of reliable, easy-to-use software that empirically finds the best algorithm for a given problem.

6 Conclusion

We have conducted a comparison of predictive performance and runtime of 31 prevalent feature selection algorithms on 50 classification problems. The results revealed a leading group of FSAs that that can be used as an initial starting point for practitioners.

- For predictive performance (Group 1): Random Forest RFE, Random Forest, Decision Tree RFE, Decision Tree, AdaBoost RFE, AdaBoost.
- For speed (Group 2): Low Variance Filter, Decision Tree, Anova, Chi Squared, Lasso, Perceptron, LassoLars.

Within these two groups, viable choices for overall performance include Decision Tree RFE and Lasso. The difference in performance of Decision Tree RFE could not be determined with significance from the majority of Group 2. Analogously, there was no significant difference between Lasso and the majority of Group 1. Ultimately, Decision Tree was found to be the best choice in terms of overall performance, since it was present in both groups.

The choice of method ultimately depends on the problem; there is no supreme method for all cases. However, with the broad selection of FSAs and datasets in this benchmark, we hope to have elucidated FSAs that are more likely to yield classifiers that reflect the underlying structures of the dataset in question.

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